TATA

TATA PROJECTS LIMITED

ENVIRONMENTAL, HEALTH AND SAFETY POLICY

Tata Projects Limited, engaged in Engineering, Procurement and Construction Projects in Infrastructure development and Third Party Inspection & Expediting Services in India & Overseas is committed to achieving EHS excellence through belief that all accidents are preventable. In pursuit of this commitment and belief, we strive to:

- Comply with applicable Legal and other requirements related to EHS.
- Provide Safe and Healthy environment for all employees, including those employed by sub-contractors, channel partners &visitors.
- Establish procedures to recognize, implement, evaluate and monitor EHS performance.
- Continual improvement of EHS performance through setting and reviewing relevant and appropriate objective & targets, prevention of adverse impacts on environment through climate change initiatives.
- Develop a safety culture through active leadership, encouraging good safety performance through Reward & Recognition schemes and imparting appropriate level of EHS training.
- Safe upkeep of all tools, equipments & machinery and conservation of natural resources.
- Communicate the policy to all persons working for or on behalf of the organization, including the interested parties and periodically review the policy for its relevance appropriate to the organization.

Our most valued assets are the Human Resources and both the management and employees share appropriate responsibility towards meeting the obligation & commitment of Safe, Healthy and Environmentally compliant workplace.

Vinayak K Deshpande
Managing Director

Date: 03-10-2011

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CODE OF SAFETY AT CONSTRUCTION SITE

1.0 INTRODUCTION

Uncompromising quality and safety as one of the prime missions, zero tolerance towards non-compliance is the order of the day at TATA PROJECTS. High risk activities such as working at heights, extensive material handling, extensive earthwork using all kinds of tools and plants etc com- pounded with other operational challenges such as low skilled labour, stringent time lines and com- petitive environment renders safety at TATA PROJECTS is of paramount importance and the safety culture extends to everyone associated with the Company.

The code of safety management at construction sites elaborated in this document shall be strictly adhered to by the CONTRACTOR. It is the responsibility of the CONTRACTOR to provide all the safety items to their personnel working at site. Responsibility for the safety of materials including the free issue by MAIN CONTRACTOR to CONTRACTOR and their personnel working at site solely lies with the CONTRACTOR and the MAIN CONTRACTOR shall in no way be responsible for any loss of property and injury/loss of life due to accident to the personnel working at site.

The CONTRACTOR shall ensure that MAIN CONTRACTOR's Environment, Health and Safety (EHS) standards and procedures requirements shall be strictly adhered to. The CONTRACTOR must ensure that their employees are competent and qualified to carry out their specific tasks. Necessary documents pertaining to the qualifications and certification must be submitted for evaluation and approval from MAIN CONTRACTOR's EHS Department.

The CONTRACTOR should arrange for all site facilities of their workers viz. First Aid, drinking water, toilets, waste bins on suitable location at the site.

On award of the contract, all CONTRACTORS will be issued a copy of MAIN CONTRACTOR's Operating Control Procedure (OCP) and Tata Standard Format (TSF). All work should be carried out in compliance with these OCPs and TSFs.

2.0 DEFINITIONS

The words and terms specified in this section or pronouns in their stead, shall for the purpose of this contract and the specifications thereof, have the meanings as given below.

2.1 MAIN CONTRACTOR shall mean Tata Projects Limited having their registered and corporate office at Mithona Towers-1, H No. 1-7-80 to 87, Opp Wesley Co. Ed Junior College, Prenderghast Road, Secunderabad – 500 003 who issued the tender and order and shall include their successors and assigns, as well as their authorized officers / representatives including consultants, if any of Tata Projects Limited.

2.2 ENGINEER-IN-CHARGE shall mean the person nominated by the MAIN CONTRACTOR from time to time and shall include those who are expressly authorized by the MAIN CONTRACTOR to act for and on his behalf for all functions pertaining to operation of the 'contract '/ 'order' and the representative of the MAIN CONTRACTOR who shall direct, supervise and be in-charge of the works for purposes of this Contract.

2.3 CONTRACTOR shall mean the successful BIDDER, an individual or firm or company whether incorporated or not, whose Bid has been accepted by the MAIN CONTRACTOR and on whom the 'Contract' or 'Order' is placed by the MAIN CONTRACTOR and shall include their heirs, legal representatives, successors and permitted assigns of such individual or persons composing such firm or unincorporated company as the case may be

2.4 'Accepting Authority' shall mean the authority retained by MAIN CONTRACTOR

2.5 Approved' and 'Approval' where ever used in this document shall mean respectively approved by and approval by the 'Accepting Authority' shall mean the authority retained by MAIN CONTRACTOR. Wherever in the contract, the words 'Approved',' Approval' , 'Subject to Approval' ,'Satisfactory', 'Considered Necessary', 'Designated', 'Directed', Prescribed', 'required', 'Equal to ' , 'Proper', 'Requested' , 'As directed' , 'Where directed', 'When directed', 'Determined by' , 'Accepted, 'Permitted', or words and phrases of like import are used, the approval, acceptance, judgments, re-quirement, permission, order or prescription, direction, etc; is understood to be a function of the 'Accepting Authority'.

2.6 CONTRACT shall mean the Letter of Intent or Work Order issued to the CONTRACTOR by the MAIN CONTRACTOR or the agreement between the MAIN CONTRACTOR and the CONTRACTOR by acceptance of CONTRACTOR's offer by MAIN CONTRACTOR for executing the works required.

2.7 'CONTRACTOR's Equipment' or 'Constructional Plant' or 'Plant, Tools & Equipment' means all appliances and things of whatsoever nature (other than temporary works) required for the execution, completion or maintenance of the Works or temporary works (as hereinafter defined) and the remedying of any defects therein, but does not include machinery, materials or other things intended to form or forming part of the permanent Works.

2.8 'CONTRACTOR's works' or 'Manufacturer's Works' shall mean and include the land and other places which are used by the Contractor or his SUB-VENDOR / SUB-CONTRACTOR / SUB- FABRICATOR for the manufacture of 'Equipment' or performing the 'Works'.

2.9 'Day' shall mean a day of 24 hours from mid-night to midnight irrespective of the number of hours worked in that day.

2.10 'Week' shall mean seven days without regard to the number of hours worked in any day in that week.

2.11 'Month' shall mean thirty calendar days without regard to the number of days worked in any day in that month

2.12 'Engineer In Charge's Instruction' shall mean any drawings and / or instructions oral and / or in writing, details, direction and explanations issued by MAIN CONTRACTOR from time to time during the Contract Period as further elaborated in this document.

2.13 'MANUFACTURER' refers to a person or firm who is the producer and supplier of material or designer and fabricator of equipment to be supplied to the Engineer or the contractor or both under the 'Contract' / 'Order'.

2.14 'Notice in Writing' or' Written Notice' shall mean a notice in written, typed or printed characters sent (unless delivered personally or otherwise proved to have been received) by registered post or Courier to the last known private or business address or registered office of the ad-dressee and shall be deemed to have been received when in the ordinary course of post or courier service it would have been delivered.

2.15 'OTHERS CONTRACTORS' shall mean other successful contractors who have been appointed by MAIN CONTRACTOR for other works other than envisaged in the contract of the CONTRACTOR

2.16 'Plant' or 'Equipment' and 'Work' or 'Works' shall mean work to be executed and services to be provided by the Contractor under the 'Contract' or "Work Order'

2.17 'Site' shall mean the actual place / places provided or made available by the MAIN CONTRAC-TOR in the 'Project' as detailed in the 'Specification' where 'Plant' and 'Equipment' are to be delivered or where work has to be executed under the Contract by the 'Contractor'

2.18 SUB-CONTRACTOR shall mean the individual or firm or company named in the 'Contract', or 'Order' undertaking a part of the work or any individual to whom a part of the 'Contract' / 'Order' has been sublet by the Contractor with the consent in writing of the Engineer and shall include their heirs, legal representatives, successors and permitted assigns

2.19 'Works' shall mean and include all works to be executed in accordance with the Contract or part thereof as the case may be and shall include all extras, additions, altered or substituted works as required for the purpose of the Contract.

2.20 'Writing' shall include any manuscript, typewritten or printed statement issued under or over signature of the MAIN CONTRACTOR with seal.

2.21 "Temporary works" shall mean all temporary works of every land required in or about the execution, completion or maintenance of the works.

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2.22 'Contract Period' shall mean the time period required for completing the execution of and passing the tests on completion of the works or any section or part thereof as stated in the contract (or as extended subsequently as per provisions made in the contract) calculated from the commencement date. The 'Contract' shall be executed as agreed between the MAIN CONTRACTOR and CONTRACTOR in the Letter of Intent / Work order / Contract.

2.23 The, "PROJECT" shall mean the place in India/abroad where the facility/plant is being set up.

2.24 'Specifications' shall mean:

(i) All the terms, directions, various technical specifications, data sheets and details attached to the contract,

(ii) Design Memoranda and such mutually agreed revisions and alterations as may be made to these documents which pertain to the method and manner of performing the work or works and the quantities and qualities of the materials to be furnished under the contract as may be amplified or modified by the MAIN CONTRACTOR during the performance of the Contract in order to provide for unforeseen conditions or in the best interest of the work or works. It shall also include the latest edition including all addends/ corrigenda of relevant Indian standard specifications and other relevant codes (BIS / corresponding International codes)

3 SAFETY REQUIREMENTS

3.1 RESPONSIBILITY OF THE CONTRACTOR

3.1.1 Before commencing the work, the CONTRACTOR shall appoint / nominate required qualified safety officer/personnel in consultation with MAIN CONTRACTOR / Engineer-in-Charge to su-pervise adoption and implementation of all safety measures including co-ordination. The qualification of Safety Officer should be as defined in Building and Other Construction Workers Act (1996) and the resume of the proposed officer/personnel should be approved by MAIN CONTRACTOR / ENGINEER-IN-CHARGE prior to deployment. For every 100 workers the CONTRACTOR must deploy a Safety Officer.

3.1.2 The Contractor shall submit a "safety plan" to the MAIN CONTRACTOR. The safety plan shall indicate in detail the measures that would be taken by the CONTRACTOR to ensure safety of men, equipment, material etc. during execution of the work. The plan shall take care to satisfy all requirements specified hereunder. MAIN CONTRACTOR shall have right to review, suggest modifications and approve the safety plan as required. MAIN CONTRACTOR's decision in this respect shall be final and the CONTRACTOR shall abide by the same.

3.1.3 The CONTRACTOR shall take all necessary safety precautions and arrange for appropriate protective gears as per the directives of MAIN CONTRACTOR or ENGINEER-IN-CHARGE to prevent loss of human lives, injuries to men engaged and any damage to property etc.

3.2 PERSONAL PROTECTIVE EQUIPMENT (PPE)

CONTRACTOR shall supply its staff and employees with Personal Protective Equipment (PPE) meeting MAIN CONTRACTOR's designated standards as given below. This shall comprise of a minimum; safety shoes, hardhat, and wrap around safety glasses. These items are to be worn at all times while at the SITE and outside of office. Additionally, personnel working at heights, 1.8 meters and above, shall use a full body harness with double Lanyard to be tied off at all time. PPE shall be maintained in good condition and any damage to be reported to Safety Officer and the replacement should be issued. The details of PPE Standards to be followed are:

- Safety helmets conforming to IS 2925 / 1984 (1990) & DGMS Approved.
- Full body adjustable harness with two front textile loops, 2 mtrs. Polyamide Lanyard and Hook PN 121 CE- EN 361:2002.
- Safety Eyewear, frameless with 99.9% UV protected scratch & impact resistant polycarbonate clear lens 'CE' marked confirming EN 166:2001.
- Safety Shoes conforming to IS 1989 part – II / 1986 (1992).
- Eye and face protection devices conforming to IS 2573 / 1986 (1991).
- 52.5 IS 6994 / 1973 part – I (1991), IS 8807 / 1978 (1991), IS 8519 / 1977 (1991).
- PPE's not covered above but found desirable for the site and nature of work shall be provided by the CONTRACTOR to the work force as per the MAIN CONTRACTOR's guidelines.

3.3 HYGIENE

CONTRACTOR shall ensure that adequate numbers of personnel are dedicated to housekeeping at the worksite and ensure that site work areas are maintained to a high standard of housekeeping, sanitation and hygiene in conformance to MAIN CONTRACTOR's instructions for Site Housekeeping, Sanitation & Hygiene.

3.4 EMERGENCY PREPAREDNESS

a) Fire Protection

A fire watch shall be assigned during the execution of hot work activity as necessary and required.

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Fire watch personnel shall be trained and certified by EHS Department of the MAIN CONTRACTOR, proof of training and certification shall be provided. Fire watch personnel will be identified by wearing Red overalls and safety helmet. CONTRACTOR shall ensure that necessary fire preventive measures are being followed and implemented. Fire fighting equipment such as extinguishers, fire hose and fire blanket shall be made available at:

- At all fuel and combustible materials storage areas
- All the entrance and emergency exits of the buildings
- On all equipment having diesel or gasoline engines.
- At all fuel and combustible materials storage areas
- All fire extinguishers are to be inspected prior to project execution.

b) Emergency Procedures

CONTRACTORS shall follow and respond in any unlikely emergency event in compliance to the MAIN CONTRACTOR's Evacuation Procedure. MAIN CONTRACTOR shall develop and implement emergency procedures. CONTRACTOR shall ensure that all its staff and personnel are fully familiar with the use and location of the essential emergency equipment such as fire extinguishers, fire exits, assembly points and plant alarms. The typical contents of such procedure are:

i. Assembly station allocations.

ii. Emergency duties and responsibilities.

iii. Instruction to the phone operators.

iv. Emergency actions to be taken.

v. Emergency assembly areas.

vi. Provisions of medical and environmental emergencies.

vii. Training and drills.

viii. Contact Details and location of First Aid facilities in case of Emergency.

ix. Location of fire extinguishers.

Above Procedure will be clearly emphasized and cascaded to all by the MAIN CONTRACTOR during Site Induction Training.

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c) First Aid Facilities

CONTRACTOR shall provide First Aid facilities equipped with basic First Aid Medicines at site for its employees. This First Aid facility shall be manned by a qualified First Aider as required. Inventory of Medicines shall be carried out on a monthly basis and a register should be maintained by the CONTRACTOR. The CONTRACTOR shall ensure that a vehicle is always present at site to meet any emergency requirement.

3.5 SAFETY MEASURES

CONTRACTOR is advised to issue all necessary safety equipments to workers and see that they are properly used by them. CONTRACTOR is advised to follow all necessary safety precaution such as providing sign boards and touching the installation with prior permission of the competent authority. All sorts of safety measures to be taken shall be deemed to form an integral part of agreement.

The CONTRACTOR shall be responsible for taking all safety precautions during the construction and keeping the site safe at all times. When the work is temporarily suspended he shall protect all construction materials, equipments and facilities from causing damage to existing property interfering with the operations of the station when it goes into services. The CONTRACTOR shall comply with all applicable provisions of the safety regulations clean up programme and other precautionary measures at site.

3.6 NEAR MISS / ACCIDENT / INCIDENT REPORTING

Reporting of near miss/accidents shall be in accordance with MAIN CONTRACTOR's TSF procedure. All accidents / incidents and near misses shall be recorded, reported to MAIN CONTRACTOR immediately and investigated by the CONTRACTOR. Generated reports must be submitted within 24 hrs of any such accident / incident or near miss.

3.7 LABOUR CAMP

The CONTRACTOR must establish adequate labour camps with proper water supply and sanitation facilities and should keep the area in a neat and hygienic condition. If female labourers are employed, cruches for their children must also be organized. In case the CONTRACTOR is not organising these facilities, the ENGINEER-IN-CHARGE reserves the right to impose a fine on each occasion and this amount will be deducted from the CONTRACTOR's bill.

Anti-larve work (e.g. Honeybee comb, Fungus etc): During execution of the works against this contract the CONTRACTOR shall be responsible for anti-larve work at the camp site and labour camps at his / their own cost.

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3.8 TOOLS AND TACKLES

CONTRACTOR shall ensure that all their tools, tackles, lifting appliances, material handling equipment, scaffolds, cradles, safety nets, ladders, equipments etc used by the CONTRACTOR shall be of safe design and construction and have been inspected and certified by a competent authority. A copy of the fitness certificates of these items shall be submitted to MAIN CONTRACTOR shall be submitted for verification and records. MAIN CONTRACTOR reserves the right to stop the use of any of the above items found unsuitable and/or not backed by appropriate Test Certificates.

a) Scaffold/ Ladder & Access Requirements

MAIN CONTRACTOR shall ensure that all necessary and essential requirements with regards to Scaffold / Ladder and Access requirements are to be followed. All scaffolding shall be erected / maintained and inspected by CONTRACTOR with prior approval of MAIN CONTRACTOR. All materials to be used during the construction shall be inspected and approved.

b) Crane and Lifting Equipment

All Crane Operators must have a valid heavy vehicle license and operating certificate from the recognized competent authority.

- No crane operation will take place without an appointed identifiable and experienced rigger.
- No crane operation shall commence without prior establishment of lift area demarcation and signage.
- Regardless of the weight or size of the load to be lifted, out-riggers must be fully extended and the crane wheels off the ground.
- The hoisting mechanism of a crane shall not be used for any purpose other than lifting vertically.
- Each crane in operation must be equipped with at least one approved dry chemical powder Fire Extinguisher.
- Crane operations are to be stopped when weather conditions are likely to cause hazard to personnel, plant or property; when the wind speed reaches 40 kmph; when the operator's vision is impaired because of fog, darkness or sandstorms.

- When the crane is being refuelled, all safety precautions such as grounding, shutting off the crane's engine, availability of fire extinguishers etc. must be ensured. No hot work should be carried out within 100 meters of radius from the refuelling area.
- Crane with more than one ton lifting capacity shall be fitted with a safe working load (SWL) indicator and a crane capacity load chart displayed inside the operators cabin. All operators should be fully familiar with load Charts and operating procedures and instructions of the crane under use.
- Cranes lifts, large and heavy loads, using the principle of leverage. Choosing the rightcrane for the job, knowing the weight and center of gravity of each load, and remaining aware of jobsite conditions are important factors toward accomplishing a job safely and efficiently.
- A safe lift begins by following the manufacturer's specifications and load capacity charts unique to each crane. For rough-terrain cranes that includes making sure the outriggers are fully extended. When set up on the ground, conditions also should be checked and monitored with each lift to ensure the forces exerted by the crane do not exceed the soil's load-bearing capacity. This is important if the crane remains on a jobsite over numerous days or if inclement weather has occurred.

CAUTION!!!

Although crane operators are licensed experts, it is important that every person on a jobsite remain conscious of the crane. A number of recommendations for riggers and ground workers are outlined in National Institute for Occupational Safety and Health (NIOSH) safety alert.

- Never work directly under a suspended load.
- Always check for overhead power lines and other obstructions.
- Barricade the swing radius to keep unauthorized persons from entering areas of pinch points.
- Follow a written engineered lift plan for all critical lifts.
- Follow the correct procedures when setting up or dismantling a crane. Make sure boom sections are blocked or supported before removing pins. Stay out from under the boom at all times, if possible.
- Know the possible pinching and crushing hazards around the machine and the load landing area.
- Be familiar with the standard crane hand signals.

c) Lifting Tackle

- All items of lifting tackle shall be marked with an identification number and Safe Working Load. The identification number should co-relate to the number indicated in the Test Certificate of the item.
- No lifting shackles shall be used beyond the period of certification and all must be free from any obvious defects. All rigging operations must be planned and supervised by a competent and trained rigger. Rigging plans must conform and meet the specifications and guidelines of the MAIN CONTRACTOR.
- Weight of the load, rated capacity of the equipment and other lifting accessories used shall be determined prior to execution of specific work activities.
- Lifting methods shall be prepared for all critical and heavy lifts.
- Tag lines must be attached to all loads that are being moved.
- Colour Coding for all lifting tools and tackle will be as determined by MAIN CONTRACTOR.

d) Hand Tools

CONTRACTOR shall ensure that all portable hand tools are inspected to ensure that they are serviceable and tested in conformance with applicable safety standards. Electric powered hand tools shall be of double insulated type or three-wire earthed rated 220 volts.

All electrical/welding cables/junction boxes shall be inspected by the MAIN CONTRACTOR's EHS Department prior to commencement of the construction activities.

3.9 MATERIAL HANDLING

CONTRACTOR shall ensure that all personnel involved in the activity have attended and satisfactorily completed the Material Handling Awareness training. Execution of work shall be in compliance to MAIN CONTRACTOR's/ENGINEER-IN CHARGE.

CONTRACTOR's Supervisors / Foremen shall ensure that material handling in their respective areas are controlled and executed in the safest possible manner without causing any hindrance to other contractors working concurrently in same / nearby area. Certified and experienced rigger / sarang shall be deployed for slinging materials. Areas within the radius of lift shall be barricaded to restrict the trespassing of unrelated personnel. A suitable and certified metallic basket shall be used when-ever loose materials and tools are being hoisted or lowered from elevated locations.

The necessary safety guards should be provided on the winches.

3.10 ELECTRICAL EQUIPMENT

All electrical equipment, connections and wiring for construction power, its distribution and use shall conform and comply with the requirements of Indian Electricity Act and rules as applicable. Only qualified licensed electricians approved by appropriate statutory authority shall be employed by the CONTRACTOR to carry out all types of electrical works. All electrical equipment including portable electric tools used by the CONTRACTOR shall have safe plugging system to power source and should be properly earthed.

Danger boards in English, Hindi / local language shall be provided at tapping points of source of power or wherever necessary. No loose wires shall be used at tapping points. Proper industrial plugs / receptacles shall be used. ELCB's of rating 30mA shall be provided for all electrical connections to ensure personal safety against electric shocks. Protective earthing shall be provided wherever necessary.

3.11 PETROLEUM / EXPLOSIVES

Where it becomes necessary to provide and / or store petroleum products, explosives, chemicals and liquid or gaseous fuel or any other inflammable substance that may cause fire or explosion, the CONTRACTOR shall be responsible for strict adherence of such provisions and / or storage in accor- dance with the rules and regulations laid down by the relevant government acts, such as petroleum act, explosives act in all such matters.

3.12 SAFE LIFTING OF EQUIPMENTS

All lifting tackles including wire ropes, slings, shackles etc. used by the CONTRACTOR shall have to be approved by ENGINEER-IN-CHARGE before they are actually put on the work. It will be the responsibility of the CONTRACTOR to ensure safe lifting of the equipment taking due precautions to avoid any accidents and damage to other equipments and personnel. All piping shall be adequately supported and protected to prevent damage during handling and erection. The history cards for major equipments are to be maintained by the CONTRACTOR.

3.13 HOISTING AND RIGGING SAFETY

CONTRACTOR should give training to their workers in the following for hoisting and rigging.

a) Sling and hitch types

b) Sling capacity determination

c) Equipment inspection, care and maintenance

d) Load weight and center of gravity determination

e) Safe lifting techniques etc

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CONTRACTOR should give training to their engineer / supervisor in the following for hoisting and rigging.

a) Identify area where hoisting and rigging techniques are used.

b) Obtain necessary equipment.

c) Establish procedure for inspection, care and maintenance of equipment.

d) Ensure workers are trained.

e) Know when hoisting and rigging techniques are necessary.

f) Ensure equipment is properly maintained.

g) Ensure workers use safe lifting techniques.

3.14 HANDLING, TRANSPORTING AND STORING GAS CYLINDERS

The CONTRACTOR should take care of the following points while handling, transporting and storing gas cylinders:

I. All concerned personnel must use PPE : GLOVES, Protective Footwear, Eye Protection

II. The correct way to move gas cylinders is to: keep it in upright position, secure and with valves facing in upper direction.

III. Use mechanical aids such as trolley for movement as far as possible

IV. Use suitable cradles, slings, clamps or other effective means when lifting the gas cylinders with a hoist or crane.

V. For short distances on even ground, the practice of 'milk – churning' (manually moving the gas cylinders) can be used only by trained personnel and should never be done in case of longer distances, uneven ground, wet or icy conditions, poor lighting, or use trolley for faster movement.

VI. All personnel involved should have completed training in handling manual cylinder movements.

VII. CAUTION!!! - Never roll cylinders along the ground

VIII. Never transport the gas cylinders with valve and pressure regulator attached or with the valve in open condition.

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IX. CAUTION!!! - Never attempt to catch a falling gas cylinder.

X. Never lift a gas cylinder by its cap, valve or guard / shroud.

XI. All compressed gas cylinders shall be marked, labelled, segregated, handled and stored in accordance with Gas Cylinder regulations - cylinder handling. Safety caps shall be fitted whenever a cylinder is not being used. Hoses shall be routed separate from electrical cables.

XII. If possible carry the gas cylinders in open vehicles or trailers. If they must be carried in closed vans / cars, ensure good ventilation of the vehicles at all times.

XIII. If in the vehicle, load compartment is not separated from the driver do not carry toxic gas cylinders. Toxic gas cylinders are marked with toxic gas label and have yellow as a colour on the cylinder.

XIV. While transportation, secure the gas cylinders properly so that they cannot move or fall in transit. Also it must be ensured that the gas cylinders should not project beyond the edges of the vehicle. Normally the gas cylinders must be in the upright position unless instructions for transport stated otherwise.

XV. It should be ensured that the gas cylinders are clearly labelled to show its contents and hazards associated with them.

XVI. The gas cylinders should be fitted with suitable protective valve caps and covers.

XVII. Always disconnect hoses and regulators from cylinders whenever the gas cylinders are to be transported.

XVIII. Always carry propane cylinders upright and do not carry flammable gas cylinders in the same compartment as toxic gas cylinders.

XIX. Do not smoke while carrying cylinders inside vans / cars.

XX. Unload the cylinders from the vehicle as soon as possible and move them to a well ventilated storage area.

XXI. In case of leakage of gas from any gas cylinder, park the vehicle, investigate the fault and contact the supplier of the gas cylinders.

XXII. If case of any accident, please inform the emergency services the type of gas cylinder being carried.

XXIII. It is best to store gas cylinders in the open and on concrete floor in a fenced compound

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XXIV. storage areas, oxygen cylinders must be stored at least 3 meters away / separated by a fire wall from fuel gas cylinders such as acetylene, propane, methane etc.

XXV. Full gas cylinders should be stored separately from empty ones. Further the empty oxygen cylinders should be segregated from empty fuel gas cylinders.

XXVI. Products like oil, corrosive liquids or any flammable materials should not be stored in the gas storage areas.

XXVII. Ease of access into and around storage area, such as provision of aisle which is greater than 0.6 m to prevent domino effect of the gas cylinders.

XXVIII. LPG cylinders have special requirements including storing them 3 mtrs. away from other gas cylinders

XXIX. Pyrophoric and toxic gases should be stored separately in locked, suitable ventilated storage areas with restricted access

XXX. Protect gas cylinders from external heat sources which can adversely affect their mechanical integrity and if possible store them away from the fire exits, elevators, walkways, building egresses, un-protected platform edges, or in locations where heavy moving objects may strike or fall on them.

XXXI. Cylinders should be secured and properly restrained by straps or chains connected to a wall bracket or other fixed surface, or by the use of a cylinder stand

XXXII. Acetylene and propane gas cylinders must never be stacked horizontally either in storage or in use.

XXXIII. Storage arrangements should ensure turnaround of stock in adequate time

XXXIV. Storage areas with gas cylinders must have more than one exit.

XXXV. Storage arrangements should be clearly described in the emergency plan displayed.

XXXVI. Storage area must have good housekeeping and adequate warning signs with fire fighting equipment as necessary.

3.15 HEALTH, SAFETY AND ENVIRONMENTAL MEETINGS

3.15.1 MAIN CONTRACTOR's EHS Personnel and CONTRACTOR's EHS representative will conduct coordination meetings to discuss topics pertinent to the nature of the present activities on site.

3.15.2 A weekly General EHS / First Aid Meeting to improve the level of communication between MAIN CONTRACTOR and CONTRACTOR shall be held. The objective of these meetings is to increase the safety awareness of the hazards, which are constantly present and to promote a keenness to put into practice the essential actions that are necessary to eliminate or control these hazards. The relevant Foreman/ Supervisor/ Safety Supervisor and EHS Manager from both sides will attend this meeting.

3.15.3 Outcome of the discussion arising from these meeting shall be recorded.

3.16 TOOLBOX TALKS

3.16.1 It is mandatory for all Supervisors / Foremen of the CONTRACTOR to conduct and deliver a daily toolbox talk at work locations within their respective area prior to commencement of daily work assignments.

3.16.2 Supervisors/ Foremen of the CONTRACTORS shall ensure that the potential hazards of the task and specific safety requirements required when executing the tasks are clearly under-stood and briefed to all the personnel involved and ensure that safe working procedures are strictly followed.

3.16.3 Toolbox Talk checklist shall include the following as applicable for the work being executed. This list is not exhaustive.

Emergency Procedures	Tools	Permit To Work (PTW)
Muster points/ alarms	Short Cuts	Lifting
PPE	Confined Space Safety	Rigging
Site Safety Regulations	Near Miss	Manual Material Handling
Housekeeping	Cylinders	Radiography
Fire Prevention	Scaffolding	Pressure tests
Safe Working Practices	Preventing Accidents	
Employees Safety Responsibility	Deep Excavations	
Eye Protection	Restricted Areas Working	

3.16.4 All workmen assigned for specific area of works shall sign the attendance sheet after completion of toolbox talks and this shall be furnished by the CONTRACTOR to the MAIN CONTRACTOR to enable them to issue the necessary work permits.

3.17 EXCAVATION / TRENCHING

Prior to any excavation the CONTRACTOR shall carry out survey of the working area to determine the existence of any underground services like electrical cables, water lines, drainage etc and obtain work permit from the MAIN CONTRACTOR for carrying out the work. The excavated spoil should not be stored within one metres of the excavation site.

Ladders or steps shall be provided for suitable access at 10 linear metre intervals. Warning barriers shall be erected around excavations with signs indicating the hazard. Warning lights shall be deployed during the hours of darkness around the excavated area. Heavy equipment shall not be operated in close proximity to the excavation site while personnel are engaged in the excavation activity.

Execution of work shall be in compliance to MAIN CONTRACTOR/Engineer-in-charge's instructions.

3.18 SCAFFOLDING

Avoid usage of wooden planks for platforms in scaffolding. The material used for scaffolding should be Steel tubes.

3.18 BLASTING

The following safety rules and regulations will be strictly adhered to:

a) Only authorized, qualified and experienced personnel shall handle explosives
b) No person shall be allowed to handle, use or work in the area while under the influence or suspicion of being under the influence of drugs or alcohol.
c) No flame, heat or spark producing device shall be permitted in ir near explosives during han-dling, transport or use.
d) Original containers shall be used to transport of detonators or explosives from storage area to blast site. Detonators and explosives should be transported in different vehicles.
e) Every reasonable precaution shall be taken to notify land owners or residents within 1000 feet of the blast site. Warning signs, with lettering a minimum of 4 inches in height on a contrasting background, will be erected and maintained at all approaches to

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the blast area. Flaggers will be stationed on all roadways passing within 1000 feet of the blast area to stop all the traffic during blasting operation. All personnel not involved in the actual detonation shall stand back at least 1000 feet and workers shall stand back from the time the blast signal is given.

f) Blasting shall be performed during daylight hours.

g) Blasting will concur within the environmental guidelines of the project and the site-specific requirements

h) Damaged or deteriorated blasting supplies shall not be used.

i) Delivery and issue of explosives shall be made by and to only authorized persons.

j) All loading and firing will be directed and supervised by a competent and experienced person.

k) Precautions will be taken to minimize the potential hazard of a premature detonation due to induced currents by using only non-electric blasting methods.

l) Blasting mats or back fill material must be utilized to control fly rock damage to surrounding structures.

3.19 PERMIT TO WORK

CONTRACTOR's personnel assigned to supervise work which requires permit shall attend a training course organized and conducted by MAIN CONTRACTOR's EHS Department. Only those persons trained and certified in accordance with the Permit to Work system will be allowed to supervise works that need a permit system. Permit to work procedure should be followed as per MAIN CONTRACTOR's safety instructions

3.20 CONFINED SPACE ENTRY

CONTRACTOR shall ensure that all personnel involved in the activity shall attend and satisfactorily complete the confined space entry training/awareness by MAIN CONTRACTOR's EHS Department.

3.21 WORK AT HEIGHT

CONTRACTOR shall ensure that all personnel involved in height work are medically fit. CONTRACTOR shall provide their personnel full body harness as per the specification mentioned in the Personal Protective Equipment guideline above and fall arrestor as per the requirement given below.

- FALL ARRESTER WITH HOOK PN 112 (C), FOR ROPE
- ANCHORAGE LINE LONG POLYAMIDE ROPE WITH ONE SIDE HOOK PN 121 & OTHER SIDE END STOP KNOT.
- CONTRACTOR shall ensure that all measures as per the MAIN CONTRACTOR OCP-Steel Erection and Work at height are followed.

3.22 GAS CUTTING

CONTRACTOR shall ensure that all precautions as laid down in OCP-Gas Cutting procedure are followed during gas cutting. No Gas Cutting shall be carried out without Flash Back arrestor. It is the responsibility of the CONTRACTOR to provide flash back arrestor. CONTRACTOR shall provide required fire extinguisher where Gas Cutting activity is in progress.

3.23 WELDING

CONTRACTOR shall ensure that welders are provided with proper PPE which includes welding shield, apron, Hand gloves, shoes etc. It is the responsibility of the CONTRACTOR to provide required fire extinguisher and the fire Extinguisher should be available near the welding activity for fire protection. CONTRACTOR shall ensure that no electrode bits are thrown on ground or platform. CONTRACTOR shall provide non flammable container to the welder for keeping the bits. CONTRACTOR shall abide by the MAIN CONTRACTOR's OCP-Welding procedure.

3.24 HOUSEKEEPING

The CONTRACTOR shall at all times ensure that:

1. Clear access shall be maintained to emergency exits at all times.
2. Clear access shall be maintained to fire fighting equipments, alarm points, confined spaces, entrance, exit and first aid kits.
3. When not in use, materials, tools, equipments should be properly stored with easy accessibility.
4. Rubbish should be removed from the work site at the end of every day.
5. Suitable bins will be provided for general site debris. Disposal of these wastes from the work site will be determined by the amount of waste accumulated. Storage of waste within the site should be minimized and for minimal duration.

6. CONTRACTOR's foreman, charge-hand or supervisors shall conduct daily inspections on their respective areas and see that they comply with good housekeeping. Verification of clean work site location prior to commencement of work activities shall be done on daily basis.

7. Access areas have to be kept free of any obstructions and any violation will be first rectified by a warning and if not cleared, then a fine of Rs. 1000/- per day will be levied till the obstructions are removed.

All wooden boards shall be promptly de-nailed upon removal and nails shall be collected and disposed of in proper containers.

3.25 WASTE MANAGEMENT / ENVIRONMENTAL PLANS

CONTRACTOR shall comply with, and ensure that all Health, Safety & Environment requirements stipulated by MAIN CONTRACTOR for disposal, storage, handling of hazardous and non-hazardous materials shall be strictly adhered to.

CONTRACTOR shall ensure that the disposal of hazardous and non-hazardous materials utilized on site is in accordance with Ministry of Environment and Forests regulations and shall obtain any necessary permits. Scrap bins / skips will be utilized for collection of the waste materials.

Manufacturer's Material Safety Data Sheets (MSDS) shall be made available at site for all hazard-ous materials, chemicals and gases.

3.26 NON DESTRUCTIVE TESTING

While using radioactive source during Non Destructive Testing (NDT), necessary safety procedures are to be followed including the statutory requirement for qualification of radioactive source techni-cian, storage and handling of radioactive source etc.

3.27 LIGHTING AND WATCH AND WARD

The CONTRACTOR shall arrange at his cost (wherever not specified) appropriate illumination at all work spots for safe working when natural day light may not be adequate for clear visibility

The CONTRACTOR shall provide and maintain at its own expenses all lighting, fencing, and watching when and where necessary for the proper execution and the protection of the facilities, or for the safety of the stored and installed equipments.

3.28 STATUTORY REQUIREMENTS AND PENALTIES FOR VIOLATION

3.28.1 The CONTRACTOR shall be held responsible for any violation of statutory regulations (local, state or central) and MAIN CONTRACTOR's instructions, which may endanger safety of men, equipment, material and environment in their scope of work or another contractors or agencies. Cost of damage, if any, to life and property arising out of

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such violation of statu-tory regulations and MAIN CONTRACTOR's instructions shall be borne by the CONTRACTOR.

3.28.2 In case of a fatal or disabling injury / accident to any person at construction sites due to lapses by the CONTRACTOR, the victim and / or his / her dependents shall be compen-sated by the CONTRACTOR as per statutory requirements. However, if considered neces-sary, MAIN CONTRACTOR shall have the right to impose appropriate financial penalty on the CONTRACTOR and recover the same from payments due to the CONTRACTOR for suitably compensating the victim and / or his / her dependents. Before imposing any such penalty, appropriate enquiry shall be held by MAIN CONTRACTOR giving opportunity to the CONTRACTOR to present his case.

3.28.3 In case of any damage to property due to lapses by the CONTRACTOR, MAIN CONTRACTOR shall have the right to recover cost of such damages from payments, due to the con-tractor after holding an appropriate enquiry.

3.28.4 In case of any delay in the completion of a job due to mishaps attributable to lapses by the CONTRACTOR, MAIN CONTRACTOR reserves the right to recover cost for such delay from payments, due to the contractor after notifying the contractor suitably and giving him opportunity to present his case.

3.28.5 If the CONTRACTOR fails to improve the standards of safety in its operation to the requirements of MAIN CONTRACTOR, after being given a reasonable opportunity to do so, and / or to provide necessary safety devices and equipment or to carry out instructions regarding safety issued by the authorized MAIN CONTRACTOR's official, MAIN CONTRACTOR reserves the right to take corrective steps at the risk and cost of the contractor after giving a notice of not less than seven days indicating the steps that would be initiated by MAIN CONTRACTOR

3.28.6 The CONTRACTOR shall submit report of all accidents including minor in nature, fires and property damage, unsafe occurrences to the authorized MAIN CONTRACTOR's official im-mediately after such occurrence but in any case not later than twelve hours of the occur-rence. Such report shall be furnished in the manner prescribed by MAIN CONTRACTOR. In addition, periodic reports on safety shall also be submitted by the CONTRACTOR to MAIN CONTRACTOR. Nonconformity of safety rules and safety appliances will be viewed seri-ously and the MAIN CONTRACTOR reserves the right to impose fines on the contractor as under:

S. No	Penalty on account of	Amount per incident in Rupees
1	Not Wearing Safety Helmet	50
2	Not Wearing Full Body Harness	100
3	Not Wearing Safety Shoe	50

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4	Not Wearing Safety Shoe	50
5	Not using 24 V supply for internal work	500
6	Not using Electric Plug	100
7	Not slinging properly	200
8	Using damaged sling	200
9	Lifting cylinder without cage	500
10	Not using proper welding cable with lot of joints and not insulated properly	200
11	Speed limits inside the plant exceeding the dis-played speed limit	500
12	Not removing scrap from platform	200
13	Gas cutting without flash back arrestor	200
14	Improper earthing of electrical T&Ps	500
15	Non Provision of 30MA ELCB	1000
16	Not using Fall arrestor while working at height	1000
17	For any Lost Time accident	10,000
18	For any Fatal accident	100,000

To enforce workforce regarding implementation of EHS norms at site, above mentioned fines will be applicable. These fines shall be imposed by MAIN CONTRACTOR's RCM & EHS Manager/incharge towards non conformity and their decision shall be binding. Any violation not included in the above list will be decided at the site and the MAIN CONTRACTOR's RCM & EHS Manager/incharge decision will be binding on that.

3.28.7 Without prejudice to the right conferred to the MAIN CONTRACTOR for stoppage of work for violations of safety requirements, the CONTRACTOR shall be liable, after writ-ten warning, for a penalty of Rs.1,000/- for the first violation and up to Rs.3,000/- for the second violation. For the third violation, the CONTRACTOR shall be liable to be de-barred from further Contracts up to a period of one year from the date of issue of debarring notice. For low value contracts up to Rs.5 crores, fines up to Rs.25,000/-, for medium value contracts from Rs.5 crores to Rs.10 crores fine up to Rs.50,000/- and for high value contract above Rs.10 crore, fine up to Rs.1,00,000/- at a time can be im-posed by MAIN CONTRACTOR on defaulting CONTRACTOR.

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3.28.8 The CONTRACTOR shall keep the MAIN CONTRACTOR completely indemnified against all losses / claims due to accidents/ injuries / damages caused at the CONTRACTOR's works site. CONTRACTOR shall be fully responsible for accidents caused due to the CONTRACTOR or his workmen's negligence or carelessness in regard to observance of the safety requirements and shall be liable to pay compensation for injuries.

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